

Grade 3 Math

Lessons, worked examples and practice

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2. Adding and subtracting with regrouping
3. Multiplication facts and strategies
4. Division as sharing and grouping
5. Fractions
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1. Place value and rounding

Every digit in a number has a place. In 4,825 the 4 is in the thousands place, the 8 is in the hundreds place, the 2 is in the tens place and the 5 is in the ones place. So 4,825 means 4 thousands + 8 hundreds + 2 tens + 5 ones, or $4,000 + 800 + 20 + 5$. This is called expanded form.

Comparing numbers

To compare two numbers, start with the biggest place. 3,462 is greater than 3,258 because the hundreds are different: 4 hundreds is more than 2 hundreds. We write $3,462 > 3,258$. The symbol always opens toward the bigger number.

Rounding

To round to the nearest ten, look at the ones digit. If it is 5 or more, round up. If it is 4 or less, round down. 67 rounds to 70. 63 rounds to 60. To round to the nearest hundred, look at the tens digit. 748 rounds to 700. 752 rounds to 800.

Example: Round 3,586 to the nearest hundred. The tens digit is 8, so round up: 3,600.

Practice

1. Write 6,539 in expanded form.
2. Which is greater, 5,204 or 5,240? Use $>$ or $<$.
3. Round 84 to the nearest ten.
4. Round 349 to the nearest hundred.
5. What is the value of the 7 in 2,703?

2. Adding and subtracting with regrouping

When you add numbers and a column adds up to 10 or more, you regroup (carry). Write the ones digit and carry the tens digit to the next column. When you subtract and the top digit is smaller than the bottom digit, you regroup (borrow) from the next place.

Example (adding): $478 + 256$. Ones: $8 + 6 = 14$, write 4 and carry 1. Tens: $7 + 5 + 1 = 13$, write 3 and carry 1. Hundreds: $4 + 2 + 1 = 7$. The answer is 734.

Example (subtracting): $602 - 245$. There are 0 tens, so borrow from the hundreds: 6 hundreds becomes 5 hundreds and 10 tens. Then borrow again to make the ones 12. $12 - 5 = 7$, $9 - 4 = 5$, $5 - 2 = 3$. The answer is 357.

Check your work

Addition and subtraction are opposites. To check $734 - 256$, you should get 478. Estimating helps too: $478 + 256$ is close to $500 + 250 = 750$, so 734 is reasonable.

Practice

1. $345 + 289 =$
2. $1,276 + 458 =$
3. $700 - 358 =$
4. $1,000 - 463 =$
5. Estimate $612 + 289$ by rounding to the nearest hundred.

3. Multiplication facts and strategies

Multiplication is repeated addition. 4×3 means 4 groups of 3, which is $3 + 3 + 3 + 3 = 12$. The numbers you multiply are called factors, and the answer is called the product. Order does not change the product: $4 \times 3 = 3 \times 4$. This is the commutative property.

Strategies that make facts easy

Times 2: double the number ($2 \times 7 = 14$). Times 5: the answer ends in 0 or 5 ($5 \times 6 = 30$). Times 10: add a zero ($10 \times 8 = 80$). Times 9: the digits of the answer add up to 9 ($9 \times 4 = 36$ and $3 + 6 = 9$). Times 4: double and double again (4×6 : $6 + 6 = 12$, $12 + 12 = 24$). Break apart a hard fact: $7 \times 6 = (5 \times 6) + (2 \times 6) = 30 + 12 = 42$.

Key facts to know by heart

$3 \times 3 = 9$, $3 \times 4 = 12$, $3 \times 7 = 21$, $4 \times 4 = 16$, $4 \times 8 = 32$, $5 \times 5 = 25$, $6 \times 6 = 36$, $6 \times 7 = 42$, $7 \times 7 = 49$, $7 \times 8 = 56$, $8 \times 8 = 64$, $9 \times 9 = 81$.

Example: There are 6 boxes with 8 crayons in each box. $6 \times 8 = 48$ crayons.

Practice

1. $7 \times 6 =$
2. $9 \times 8 =$
3. $4 \times 12 =$
4. Use breaking apart to solve 8×7 .
5. There are 5 rows of 9 chairs. How many chairs are there?

4. Division as sharing and grouping

Division splits a number into equal groups. $12 \div 3 = 4$ can mean sharing 12 things among 3 people so each gets 4, or making groups of 3 from 12 things to get 4 groups. The number you divide is the dividend, the number you divide by is the divisor, and the answer is the quotient.

Multiplication and division are partners

Because $4 \times 3 = 12$, we know $12 \div 3 = 4$ and $12 \div 4 = 3$. These four facts belong to one fact family: $3 \times 4 = 12$, $4 \times 3 = 12$, $12 \div 3 = 4$, $12 \div 4 = 3$. Use a multiplication fact you know to find the missing number: for $56 \div 7$, ask 7 times what is 56? The answer is 8.

Remainders

Sometimes things do not share out evenly. $14 \div 4 = 3$ with 2 left over, written 3 R2, because $4 \times 3 = 12$ and $14 - 12 = 2$. The remainder is always smaller than the divisor.

Example: 25 cookies are shared by 6 children. $6 \times 4 = 24$, so each child gets 4 cookies and there is 1 cookie left over (4 R1).

Practice

1. $36 \div 6 =$
2. $63 \div 9 =$
3. Write the fact family for 5, 6 and 30.
4. $29 \div 5 = ?$ (include the remainder)
5. 48 pencils are put into bags of 8. How many bags are filled?

5. Fractions

A fraction names equal parts of a whole. In $\frac{3}{4}$ the bottom number (denominator) tells how many equal parts the whole is cut into, and the top number (numerator) tells how many parts we are talking about. $\frac{3}{4}$ means 3 of 4 equal parts.

Unit fractions and comparing

A unit fraction has 1 on top, like $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$ or $\frac{1}{8}$. The bigger the denominator, the smaller each part: $\frac{1}{8}$ is smaller than $\frac{1}{4}$. When the denominators are the same, the bigger numerator is the bigger fraction: $\frac{5}{8} > \frac{3}{8}$. When the numerators are the same, the smaller denominator is the bigger fraction: $\frac{1}{3} > \frac{1}{5}$.

Equivalent fractions

Different fractions can name the same amount. $\frac{1}{2} = \frac{2}{4} = \frac{4}{8}$. On a number line they sit at the same point. Every whole number can be written as a fraction too: $1 = \frac{4}{4}$ and $2 = \frac{6}{3}$.

Example: Mia ate $\frac{2}{8}$ of a pizza and Leo ate $\frac{1}{4}$ of the same pizza. $\frac{1}{4} = \frac{2}{8}$, so they ate the same amount.

Practice

1. What fraction of the shapes is shaded if 3 of 5 equal shapes are shaded?
2. Which is greater, $\frac{3}{8}$ or $\frac{5}{8}$?
3. Which is greater, $\frac{1}{4}$ or $\frac{1}{6}$?
4. Write a fraction equivalent to $\frac{1}{2}$ with denominator 6.
5. Write 3 as a fraction with denominator 2.

6. Perimeter and area

Perimeter is the distance around a shape. Add up the lengths of all the sides. A rectangle that is 7 cm long and 3 cm wide has a perimeter of $7 + 3 + 7 + 3 = 20$ cm. Area is the amount of flat space inside a shape, measured in square units. Count the unit squares that cover the shape.

Area of a rectangle

Multiply the length by the width. A rectangle 6 units long and 4 units wide has an area of $6 \times 4 = 24$ square units. We write square centimeters as sq cm or cm squared. Area uses multiplication because the rows of squares are equal groups.

Example: A garden is 8 m long and 5 m wide. Area = $8 \times 5 = 40$ square meters. To fence it, find the perimeter: $8 + 5 + 8 + 5 = 26$ meters.

Splitting a shape

An L-shaped room can be split into two rectangles. Find each area and add them together.

Practice

1. Find the perimeter of a square with sides of 9 cm.
2. Find the area of a rectangle 7 cm by 6 cm.
3. A rectangle has an area of 36 square units and a width of 4 units. What is its length?
4. A triangle has sides 5 cm, 6 cm and 7 cm. What is its perimeter?
5. Is a perimeter measured in square units or in units? Explain.

7. Telling time and money

On a clock the short hand shows the hour and the long hand shows the minutes. Each number on the clock face is 5 minutes. When the long hand points to 6, it is 30 minutes past, or half past. When it points to 3, it is quarter past. When it points to 9, it is quarter to the next hour.

Elapsed time

Elapsed time is how much time passes between a start and an end. If a movie starts at 2:15 and ends at 4:00, count up: 2:15 to 3:15 is 1 hour, and 3:15 to 4:00 is 45 minutes. The movie lasts 1 hour 45 minutes. There are 60 minutes in 1 hour, and 24 hours in 1 day.

Money

A penny is 1 cent, a nickel is 5 cents, a dime is 10 cents and a quarter is 25 cents. 100 cents make 1 dollar. To count coins, start with the biggest value and count on. A quarter, 2 dimes and a nickel is 25, 35, 45, 50 cents.

Example: Sam buys a pencil for 65 cents and pays with a dollar. Change = $100 - 65 = 35$ cents.

Practice

1. A class begins at 9:20 and ends at 10:05. How long is it?
2. How many cents are 3 quarters and 2 dimes?
3. A toy costs 2 dollars and 45 cents. You pay with 5 dollars. How much change do you get?
4. What time is it 30 minutes after 7:45?

8. Word problems

Use four steps. 1: Read the problem and say it in your own words. 2: Decide what the question asks. 3: Choose the operation and solve. 4: Check that the answer makes sense and write a sentence with units.

Clue words

Words like in all, total, together and sum usually mean add. Words like left, difference, how many more and fewer usually mean subtract. Words like each, every, groups of and times usually mean multiply. Words like shared equally, split and per usually mean divide. Always think about the story, not just the clue word.

Example (two steps): A store has 4 shelves with 12 books on each shelf. It sells 17 books. How many books are left? Step 1: $4 \times 12 = 48$ books. Step 2: $48 - 17 = 31$ books left.

Practice

1. Ben reads 15 pages each day for 6 days. How many pages does he read in all?
2. There are 72 students going on a trip. Each bus holds 9 students. How many buses are needed?
3. Ana has 3 packs of 8 stickers. She gives 5 stickers to a friend. How many stickers does she have now?
4. A rope is 90 cm long. It is cut into 6 equal pieces. How long is each piece?

9. Answer key

1. Place value

6,000 + 500 + 30 + 9; 5,240 is greater ($5,204 < 5,240$); 80; 300; 700

2. Adding and subtracting

634; 1,734; 342; 537; about 900 ($600 + 300$)

3. Multiplication

42; 72; 48; $8 \times 7 = (5 \times 7) + (3 \times 7) = 35 + 21 = 56$; 45 chairs

4. Division

6; 7; $5 \times 6 = 30$, $6 \times 5 = 30$, $30 / 5 = 6$, $30 / 6 = 5$; 5 R4; 6 bags

5. Fractions

$3/5$; $5/8$; $1/4$; $3/6$; $6/2$

6. Perimeter and area

36 cm; 42 square cm; 9 units; 18 cm; In units, because it is a distance around the shape

7. Time and money

45 minutes; 95 cents; 2 dollars and 55 cents; 8:15

8. Word problems

90 pages; 8 buses; 19 stickers ($24 - 5$); 15 cm